

Power BI Assignment 2 – DAX & Data Visualization

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Course: Program In AI Driven Data Analytics

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DAX Calculations

Calculated Columns

Create a Calculated Column for 'Category Type

1 Category Type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]							
Amount	Profit	Profit Status	Profit Margin	Quantity	Category	Sub-Category	Category Type
₹ 12	1	Profit	0.0833333333333333	2	Clothing	Hankerchief	Clothing - Hankerchief
₹ 257	23	Profit	0.0894941634241245	5	Clothing	Hankerchief	Clothing - Hankerchief

Calculate Revenue per Order in Order Details Table:

Revenue per Order		\$ Format Currency		Σ Summarization Sum			
Fixed decimal num...		\$ % 0.00 Auto		Data category Uncategorized			
Structure		Formatting		Properties			
1 Revenue per Order = 'Order Details'[Amount] * 'Order Details'[Quantity]							
ID	Amount	Profit	Profit Status	Profit Margin	Quantity	Category	Sub

Create a Calculated Column to Categorize Sales

Text		\$ % 0.00 Auto		Data category Uncategorized		Sort by column		Data groups		Manage relationships		New column	
Structure		Formatting		Properties		Sort		Groups		Relationships		Calculations	
1 Sales Category = IF('Order Details'[Amount] > AVERAGE('Order Details'[Amount]), "Above Average", "Below Average")													
Amount		Profit		Profit Status		Profit Margin		Quantity		Category		Sub-Category	
₹ 12		1		Profit		0.0833333333333333		2		Clothing		Hankerchief	
										Clothing - Hankerchief			
												₹ 24	

Calculated Measures

Order Count- Created the measure that calculates the total number of orders in the Order Details table.

Structure	Formatting
1 Order Count = COUNT('Order Details'[Order ID])	

Created the measure that calculates the average profit for orders placed in Delhi.

Structure	Formatting	Properties	measure measure Calculations
1 Average Profit in Delhi = CALCULATE(AVERAGE('Order Details'[Profit]), 'List of Orders'[City] = "Delhi")			

Created the measure that calculates total sales accumulated from the beginning of the year up to the current order date.

List of Orders	\$ % 0.00 Auto	Structure	Formatting	Properties
1 YTD Sales =TOTALYTD(SUM('Order Details'[Amount]), 'List of Orders'[Order Date])				

Data Visualization

Sales Target Achievement by Category:

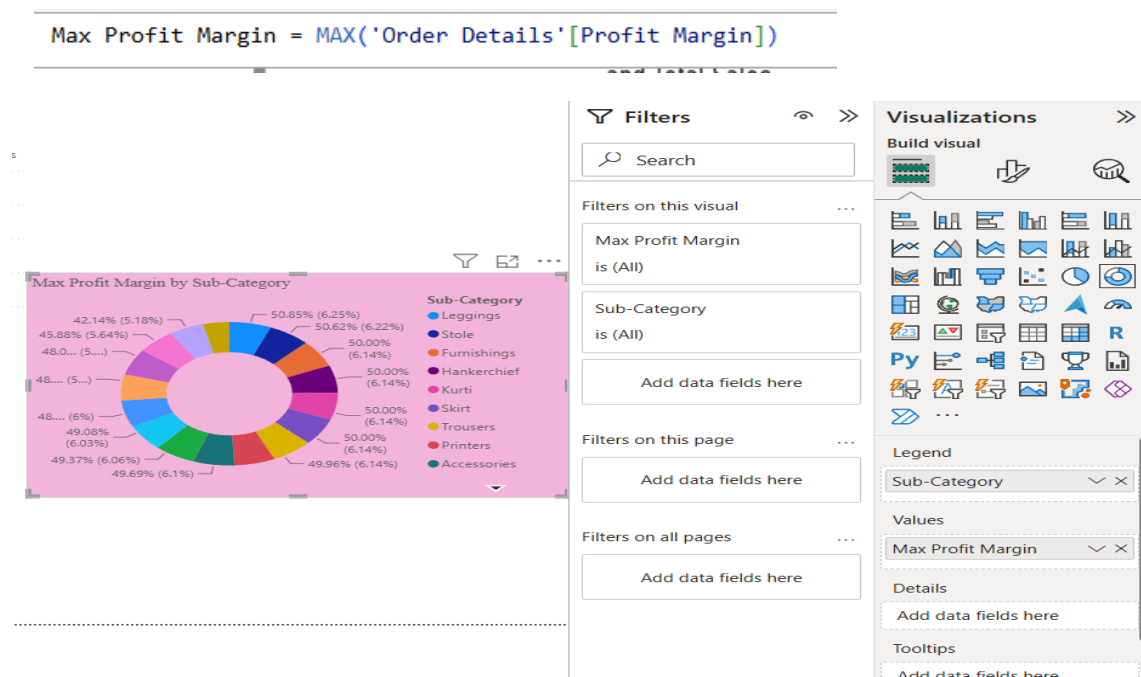
1) Created the Total Sales measure and the Total Target measure

Structure	Formatting	
Port view	Total Sales = <code>SUM('Orders Data'[Order Details.Amount])</code>	
Structure	Formatting	Pro
1	Total Target = <code>SUM('Sales Target'[Target])</code>	

using a clustered column chart Compared actual sales with sales targets by category



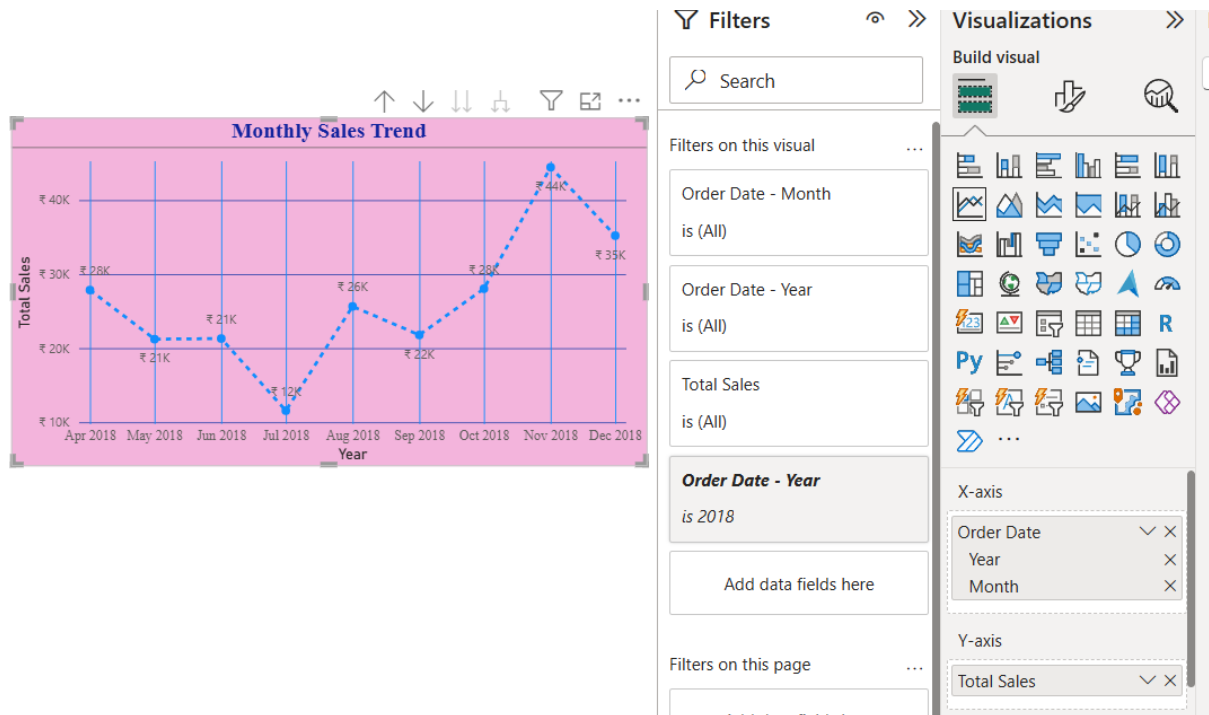
2) Max Profit Margin by Sub-Category: to Analyse the maximum profit margin 1st created the measure for max profit margin & created the Donut chart for each sub-category



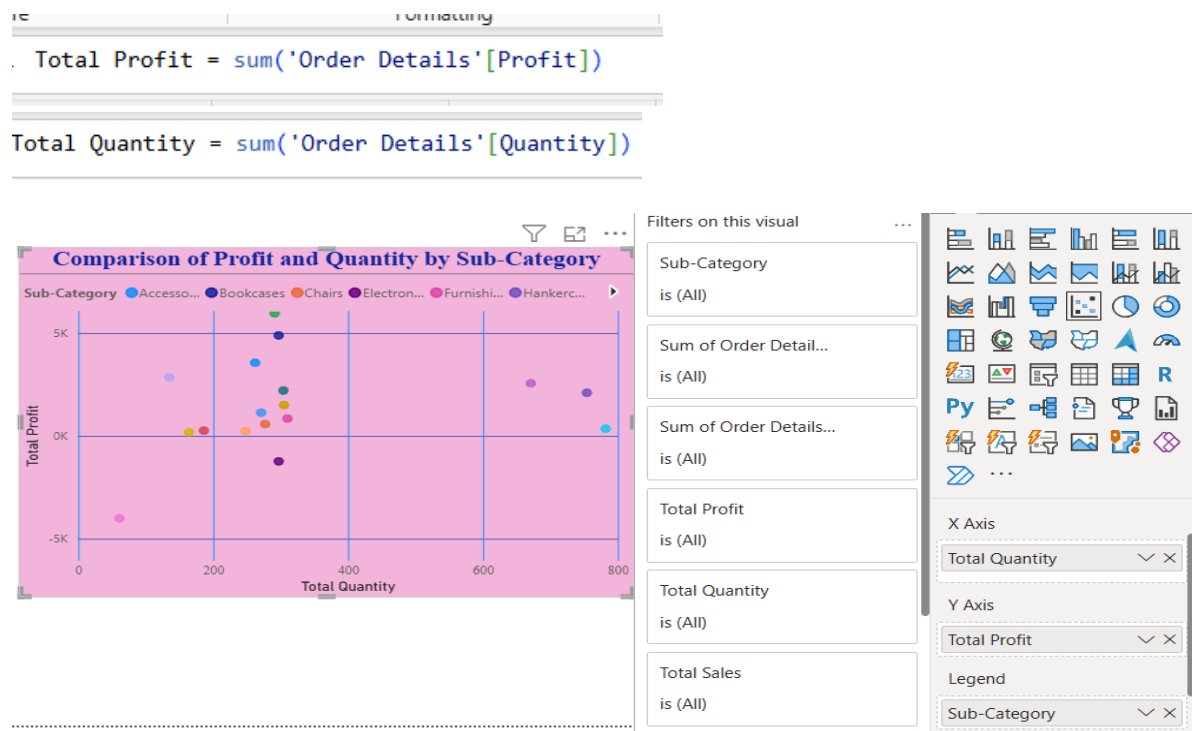
3) Monthly Sales Trend: Created the Total sales in measures & Implement the line chart for the trend of monthly sales over time

Structure	Formatting
1	Total Sales = SUM('Orders Data'[Order Details.Amount])

Insert the line charts



4) Comparison of Profit and Quantity by Sub-Category: created the measures for profit and Quantity & using the Scatter Chart compared different sub-categories.



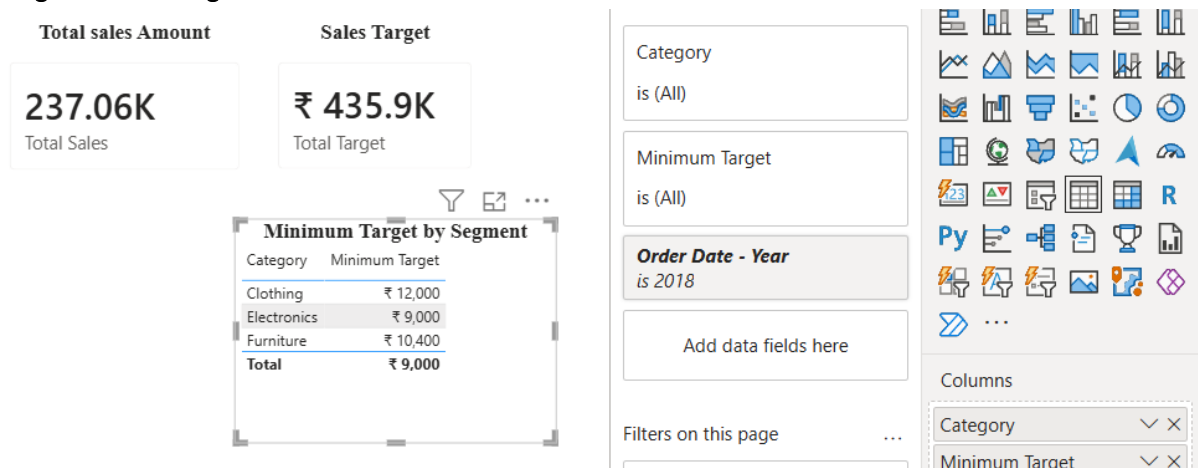
5) Comparison of Total Sales Amount and Target: Created cards to display the total sales amount alongside the sales target in Visualization.



To create a multi-row card to display the minimum target for each segment. First created the measure for Sales Target as minimum target

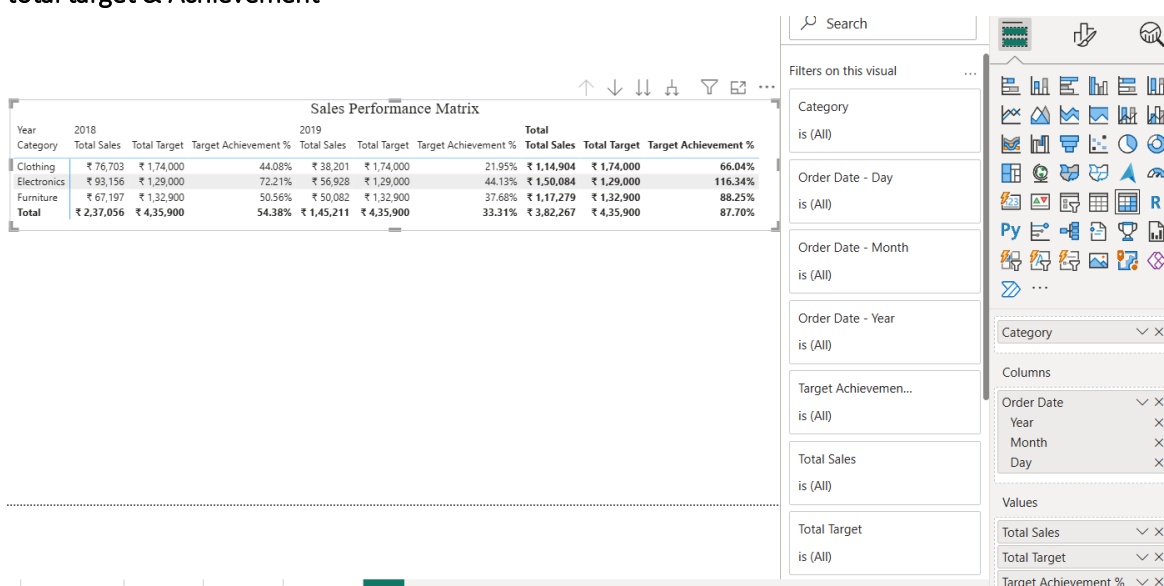
```
Minimum Target = MIN('Sales target'[Target])
```

I don't have a Multi Row card in power BI Visuals so I have used the Table Visual to display the Minimum Target for each Segment

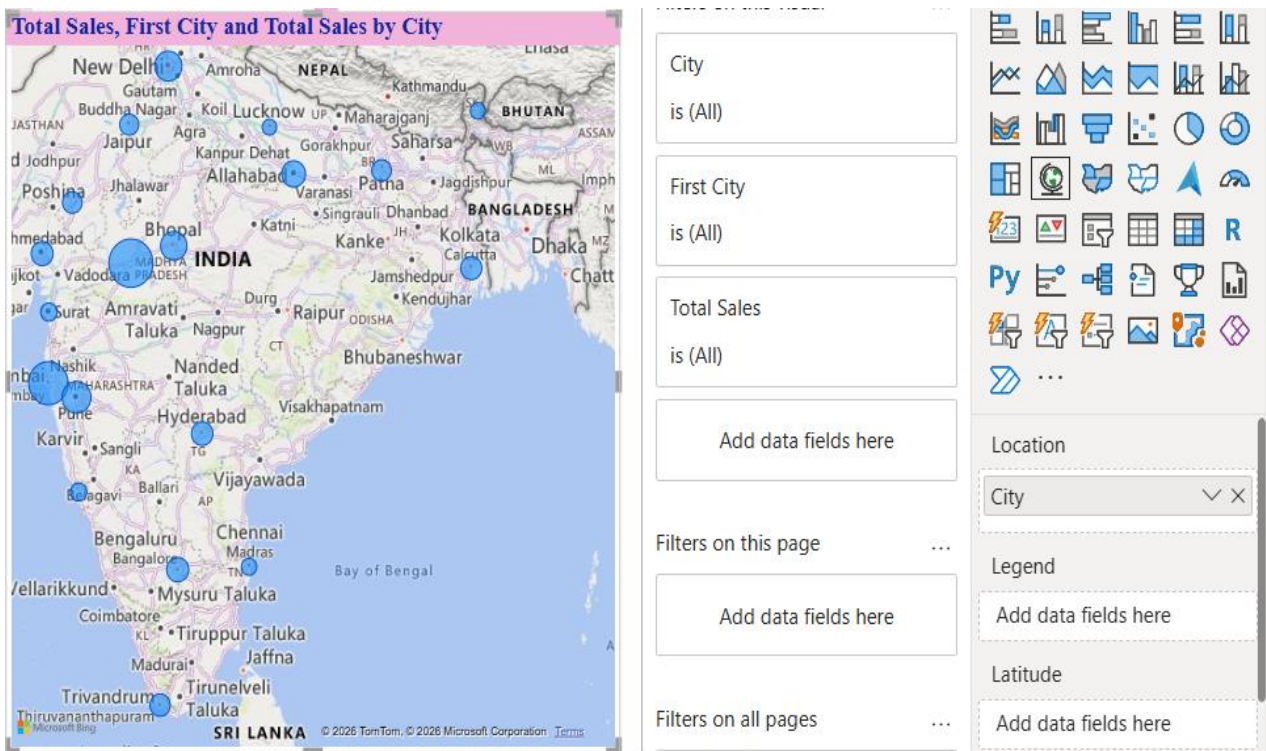


6) Sales Performance Matrix: I have created the Measure for the Target Achievement and then implement the Matrix table rows as the Category columns as Order date and values as total sales and total target & Achievement

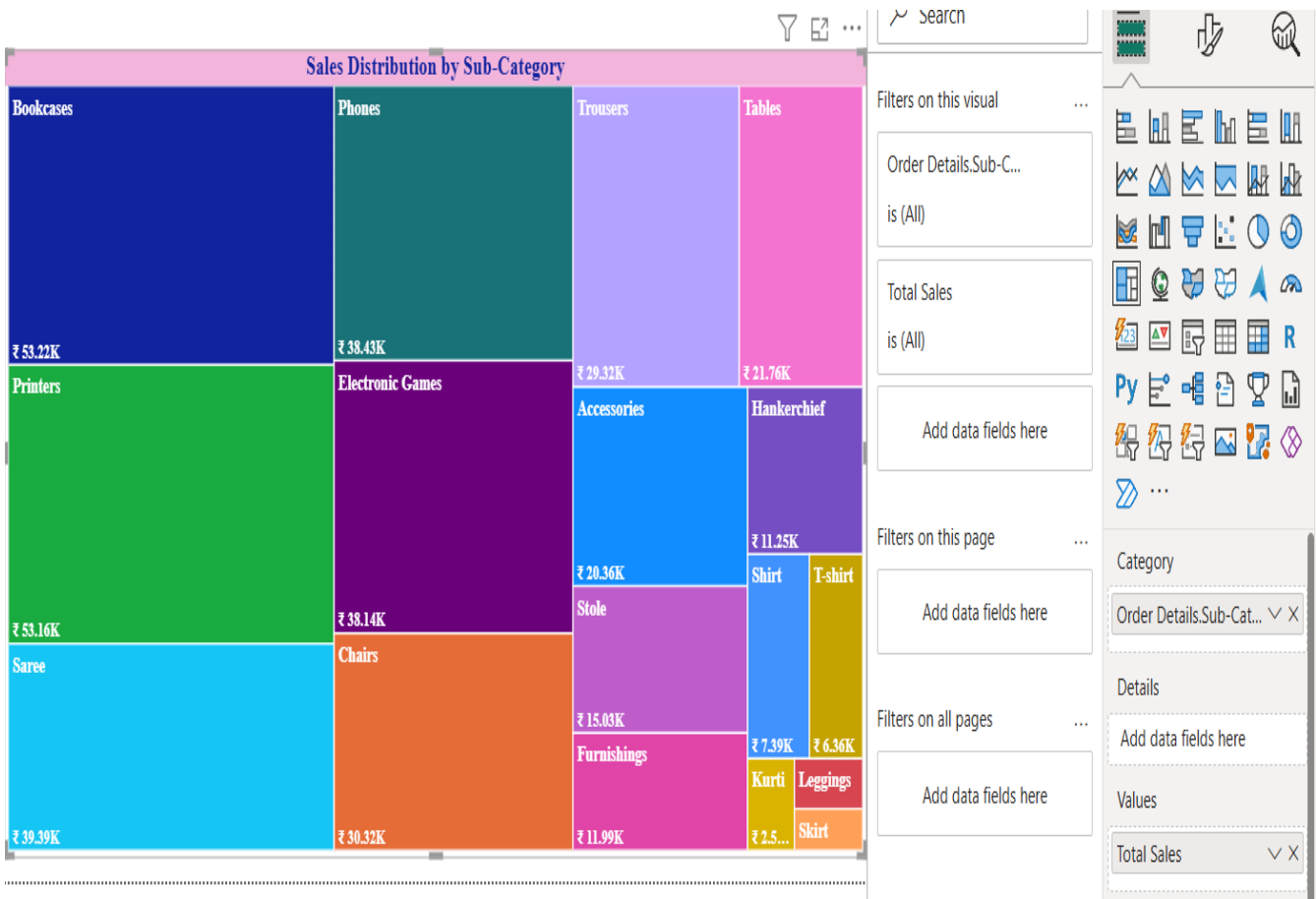
```
Target Achievement % = DIVIDE([Total Sales], [Total Target], 0)
```



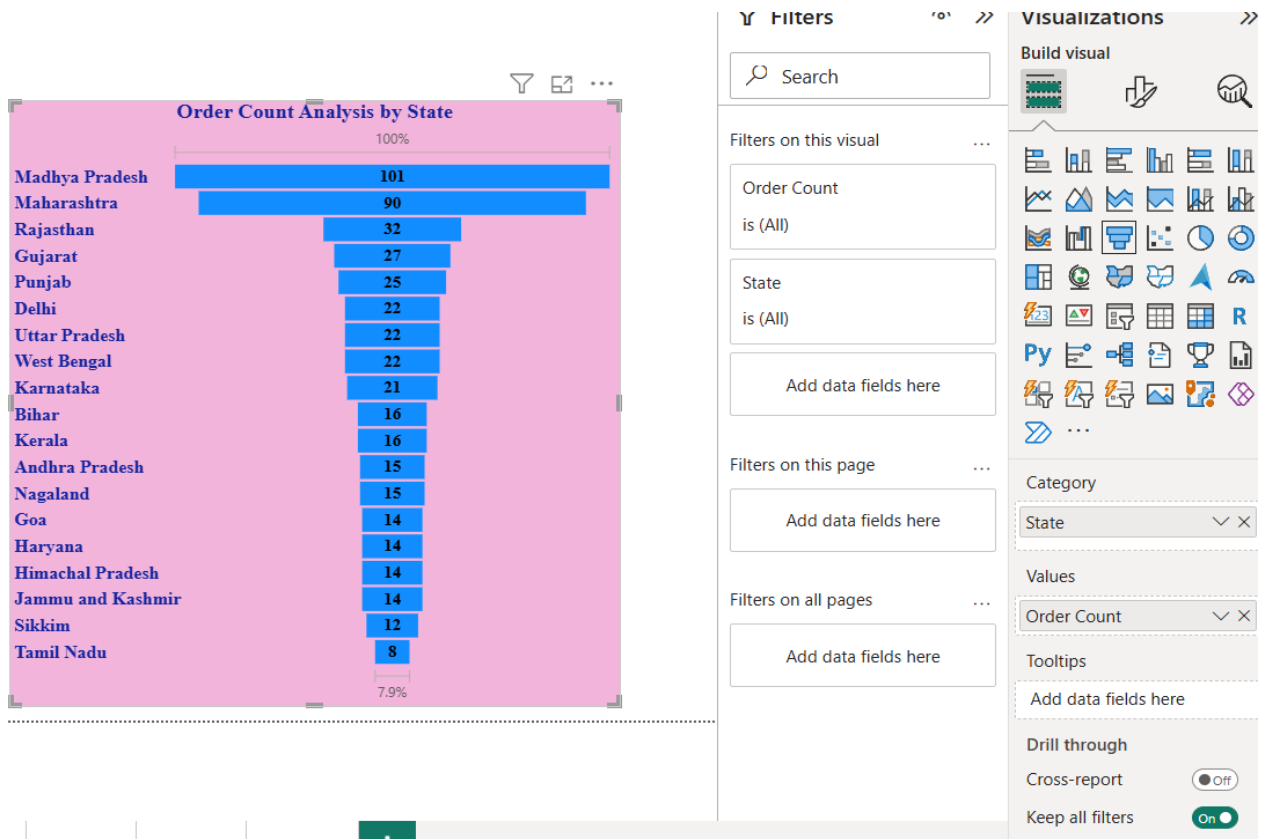
7) Geographic Sales Analysis: Visualize total sales on a map by city to identify regional sales patterns.



8) Sales Distribution by Sub-Category: created the tree map visual and category as sub-category and values as total sales



9) Order Count Analysis by State: created the funnel chart to Visualize the state category by total order count.



Dashboards :

